Factors to consider for Estimators, CIs, and HTs	**Population Correlation**
Sample Correlation	**Properties of Sample Correlation**
Simple Linear Regression (SLR) Model	**Four Assumptions about data for SLR Model**
Distribution of Y_i in SLR	**MLEs of β_0 and β_1**
Fitted Values	**Residuals**

A measure of the strength of the linear relationship between two quantitative variables in a population of interest

Method	**Consideration 1**	**Consideration 2**
Estimator	Unbiased	Minimum Variance
CI	Proper Coverage	Minimum Width
HT	Proper Size	Highest Power

- Always between -1 and 1
- Strongly influenced by outliers
- It only measures linear association, so the sample correlation could be 0 even in the presence of a strong relationship between x and y

A statistic used to estimate the population correlation

1. The observations are independent
2. The mean of Y_i is a linear function of x_i
3. The Y_i have a common variance, σ^2, which is the same for every value of x
4. The errors (and hence the Y_i) are normally distributed

$$Y_i = \beta_0 + \beta_1 x_i + \epsilon_i \quad \text{for} \quad i = 1, 2, \cdots, n$$

where ϵ_i is the random error associated with each observation such that each is iid according to a normal distribution such that $\epsilon_i \sim N(0, \sigma^2)$

- $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$
- $\hat{\beta}_1 = S_{XY}/S_{XX}$

$$Y_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2)$$

$$e_i = y_i - \hat{y}_i = y_i - \left(\hat{\beta}_0 + \hat{\beta}_1 x_i\right)$$

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

Least Squares Criteria	**Estimator for σ^2**
Interpretation of Slope Estimate	**Interpretation of Intercept Estimate**
Sampling Distribution of $\hat{\beta}_1$	**Sampling Distribution of $\hat{\beta}_0$**
CIs for β_0 and β_1	**Sum of Squares Relationship**
SST	**SSR/SSreg**

$$\hat{\sigma}^2 = s^2 = \frac{\sum e_i^2}{n-2} = \frac{RSS}{n-2} = \frac{\sum \left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i\right)^2}{n-2}$$

$$Q = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} \left(y_i - \left(\hat{\beta}_0 + \hat{\beta}_1 x_i\right)\right)^2$$

On average, the value of y is $\hat{\beta}_0$ when x is equal to 0

For every one unit increase in x, we get an average increase of $\hat{\beta}_1$ in y

$$\hat{\beta}_0 \sim N\left(\beta_0, \sigma^2\left[\frac{1}{n} + \frac{\bar{x}^2}{S_{XX}}\right]\right)$$

$$\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{S_{XX}}\right)$$

$$\text{SST} = \text{SSreg} + \text{RSS} = \text{SSR} + \text{SSE}$$

- $\hat{\beta}_0 \pm t_{n-2,1-\alpha/2}\text{se}\left(\hat{\beta}_0\right)$
- $\hat{\beta}_1 \pm t_{n-2,1-\alpha/2}\text{se}\left(\hat{\beta}_1\right)$

- $\text{SSR} = \text{SSreg} = \sum\left(\hat{y}_i - \bar{y}\right)^2$
- Variability explained by the model
- How each fitted value differed from the mean

- $\text{SST} = \sum\left(y_i - \bar{y}\right)^2$
- Total variability in the observed values of y
- How each observed value differs from the mean

SSE/RSS	**Mean Square**
ANOVA table for SLR	**Coefficient of Determination**
Dummy Variable	**Least Absolute Value (LAV) Regression**
M-Estimators	**Minimizing M-Estimators**
Iteratively Reweighted Least Squares (IRLS)	**SLR Assumptions**

The sum of squares divided by its degrees of freedom

- SSE = RSS = $\sum (y_i - \hat{y}_i)^2$
- Variability NOT explained by the model
- How each observed value differs from the fitted line

- $R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}}$
- Represents the proportion of variance in y that is accounted for by regression on x.
- It is always a value between 0 and 1
- Can be obtained by squared the sample correlation coefficient r

Source	df	SS	MS	F
Regression	1	SSR	$MSR = \frac{SSR}{1}$	$\frac{MSR}{MSE}$
Error	$n-2$	SSE	$MSE = \frac{SSE}{n-2}$	
Total	$n-1$	SST		

Estimates of β_0 and β_1 are found by obtaining the values that numerically minimize

$$\min_{\beta_0,\beta_1} \sum_{i=1} |y_i - \hat{y}_i|$$

- Good at catching outliers in vertical direction
- Bad at catching outliers in horizontal direction

- A new variable created from a categorical variable
- Will always be one fewer dummy variables than number of categories in categorical variable
- Example: Categorical Variable - New or Existing

$$x = \begin{cases} 1, & \text{if New} \\ 0, & \text{if Existing} \end{cases}$$

Instead of taking derivative of $\rho(\cdot)$, use a function on the weights that decreases as the size of the residual increases,

$$\sum_{i=1}^{n} w\left(e_i/\hat{\sigma}_e\right) \boldsymbol{x}_i = 0$$

The issue with minimizing $\sum_{i=1}^{n} \rho(e_i)$ is that we don't have an initial value of e_i. Iteratively reweighted least squares (IRLS) solves this problem.

Estimates of β_0 and β_1 are the values that minimize

$$\sum_{i=1}^{n} \rho(y_i - \hat{y}_i) = \sum_{i=1}^{n} \rho(e_i)$$

where $\rho(\cdot)$ is a function of the residuals that does not increase as rapidly as the square function does.

1. The observations are independent
2. The mean of Y_i is a linear function of x_i
3. The Y_i have a common variance σ^2, which is the same for every value of x
4. The errors (and hence the Y_i) are normally distributed

1. Get pilot estimates of β_0 and β_1 (usually with LS estimators). Denote $\hat{\boldsymbol{\beta}}^{(0)} = \left(\hat{\beta}_0^{(0)}, \hat{\beta}_1^{(0)}\right)^\top$
2. Obtain residuals based on $\hat{\boldsymbol{\beta}}^{(0)}$ (Denoted $e_i^{(0)}$)
3. Choose weight function & get preliminary weights (Denoted $w\left(e_i^{(0)}\right), i = 1, \cdots, n$)
4. Estimate β_0 & β_1 by minimizing $Q = \sum w\left(e_i^{(0)}\right)$
5. Obtain new residuals and new weights
6. Repeat steps 4 and 5 until convergence

Checking Independence Assumption	**Remedies for Autocorrelation**
Checking Linear Relationship Assumption	**Outlier**
Checking Constant Variance (Homoscedasticity) Assumption	**Checking Normally Distributed Errors Assumption**
NOTE:	**Types of Outliers**
Leverage Points	**Bad Leverage Point**

- For positive serial correlation, you can add lags of the dependent and/or independent variable to the model
- For negative serial correlation, check to make sure that none of your variables are overdifferenced
- For seasonal correlation, you can add seasonal dummy variables to the model

Use the Durbin-Watson test (test stat ranges from 0 to 4)

- $d = 2$ indicates no autocorrelation
- $d < 2$ indicates positive serial correlation
- $d > 2$ indicates negative serial correlation

If d is less than 1.5 or greater than 2.5, there is evidence of autocorrelation

- In small to moderate datasets, an outlier in a linear regression model is flagged as any point whose standardized residual falls outside of the range from -2 to 2.
- In large datasets, the accepted range is increased to -4 to 4.

- First plot the data with a scatterplot and check for a linear relationship
- Plot the residuals (y) versus the fitted values (x) from the regression model

- Create a normal QQ-plot of the standardized residuals
- Conduct the Shapiro-Wilks test for normality on the standardized residuals

Plot the square root of the absolute standardized residuals (y) against the fitted values (x) and check for any nonconstant trend

- Outlier in X Direction: high leverage; low residual
- Outlier in Y Direction: low leverage; high residual
- Outlier in X and Y Directions: high leverage, high residual

The most critical assumptions are homoscedasticity and independent observations. Without these, the variances derived for $\hat{\beta}_0$ and $\hat{\beta}_1$ do not hold

- A leverage point whose y-value does not follow the pattern set by other data points
- An outlier

- Data points that exercise considerable influence on the fitted model. Specifically, they are those points whose x-value is distant from the other x-values.
- Leverage points can either be “good” or “bad”

Good Leverage Point	**Leverage Rule of Thumb**
NOTE:	**Cook's Distance**
Relation between Cook's Distance and Leverage	**Box-Cox Power Transformations**
NOTE:	**Reasons to use Weighted Least Squares (WLS)**
SLR Model via WLS	**How do the weights, w_i, affect the variance?**

$$\sum h_i = k + 1$$

Where h_i is a single leverage point and k is the number of explanatory variables.

Leverage values greater than $2(k+1)/n$ or $3(k+1)/n$ are regarded as high.

- A leverage point whose y-value does follow the pattern set by other data points
- Not an outlier

A measure of the influence that the ith observation has on the least squares fit of a regression model, defined by

$$D_i = \frac{\sum_{j=1}^{n} \left(\hat{y}_{j(i)} - \hat{y}_j\right)^2}{2\hat{\sigma}^2}$$

where $\hat{y}_{j(i)}$ denotes the jth fitted value based on the fit obtained when the ith case has been deleted from the fit

Good leverage points still need to be addressed because they inflate R^2

- A systematic method for choosing a power transformation, λ, for a strictly positive response
- The set of power transformations is

$$Y^{(\lambda)} = \begin{cases} \frac{(Y^{\lambda}-1)}{\lambda}, & \lambda \neq 0 \\ \log(Y), & \lambda = 0 \end{cases}$$

It can be shown that

$$D_i = \frac{r_i^2}{2} \frac{h_i}{(1-h_i)}$$

so large values of Cook's distance can occur when either the standardized residual for the ith point is large, the leverage for the ith point is large, or both.

1. Stabilize Variance (Dealing with heteroscedasticity)
2. Robust Regression
3. Gives more weight to observations with more information in them

- Regression through the origin by removing the intercept should only be forced if there is a compelling reason to do so, such as data has been observed near the origin
- If no compelling reason exists and the line is forced through the origin, it may make the regression worse

- Large w_i implies (x_i, y_i) is important/has a lot of information and we want the fitted line to care a lot about this point
- Small w_i implies the opposite: (x_i, y_i) is less important

$$Y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \qquad \epsilon \overset{iid}{\sim} N(0, \sigma^2/w_i)$$

Steps for WLS	Choosing Weights

- The choice for weights is not always obvious
- An important special case is when Y_i is the average or median of n_i observations. Then the variance of Y_i is taken to be proportional to $1/n_i$, and w_i is taken to be n_i

1. Get pilot estimates for β_0 and β_1
2. Use these to get pilot residuals
3. Use pilot residuals to get pilot weights
4. Use weights in WLS to get updated estimates of β_0 and β_1
5. Repeat steps 2-4 til convergence of weights